

Lesson 2



$$\begin{array}{l}
 y = X\beta + u \quad u \sim (0, \sigma^2 I_n) \\
 \left. \begin{array}{l}
 n \times 1 \quad n \times k \times 1 \quad n \times 1 \\
 X \text{ is fixed } u \sim N(0, \sigma^2 I) \\
 X \text{ is } \rightarrow \text{stochastic } u \sim (0, \sigma^2 I) \\
 \text{rank}(X) = k
 \end{array} \right\} \text{sphericity}
 \end{array}$$

$$b = (X'X)^{-1} X'y$$

$$E(u) = 0$$

$$E(X'u) = 0 \quad \text{orthogonality condition}$$

$$E(X'u) \neq 0$$

$$\begin{array}{l}
 z \quad P \geq k \quad \text{instrument} \quad E(z'u) = 0 \\
 n \times p
 \end{array}$$

$$\min ||z'u - E(z'u)||$$

$$\begin{aligned}
 \min ||z'u|| &\sim \min u'z (Var(z'u))^{-1} z'u \\
 &\Rightarrow \frac{1}{\sigma^2} u'z (z'z)^{-1} z'u \\
 &\Rightarrow (y - X\beta)' z (z'z)^{-1} z' (y - X\beta)
 \end{aligned}$$

$$\min_b (y - Xb)' Z (Z'Z)^{-1} Z' (y - Xb)$$

$$\min_b (y' - b'X') P_Z (y - Xb)$$

$$\min_b (y' P_Z y - \underbrace{b'X' P_Z y - y' P_Z Xb + b'X' P_Z Xb}_{-2b'X' P_Z y})$$

$$-2b'X' P_Z y + 2X' P_Z Xb = 0$$

$$X' P_Z y = (X' P_Z X) b$$

$$b = (X' P_Z X)^{-1} X' P_Z y \quad P > K$$

$$K = P \quad Z \text{ is } n \times K \Rightarrow \text{rank}(Z) = K$$

$$= (X' Z (Z'Z)^{-1} Z' X)^{-1} X' Z (Z'Z)^{-1} Z' y \rightarrow \text{generalized IV}$$

$$= (Z' X)^{-1} (Z' Z) X' (X' Z)^{-1} X' Z (Z' Z)^{-1} Z' y$$

$$= (Z' X)^{-1} Z' y \quad \text{biv} \rightarrow \text{simple IV}$$

$$y = X\beta + u \quad u \sim (0, \sigma^2 I_n)$$

$$y = X\beta + u \quad E(x'u) = 0$$

$$\text{Var}(u) = \sigma^2 I$$

$$E(x'u) = 0 \rightarrow E(\beta'u) = 0$$

$$\text{Var}(u) = \sigma^2 I \rightarrow b_{IV}$$

$$\text{If } \text{Var}(u) \neq \sigma^2 I_n$$

$$= \sigma^2 \Omega_n \quad \Omega \neq I_n$$

$$\text{if } X \text{ are fixed } E(x'u) = 0$$

Generalized least square

$$\text{When both } \text{Var}(u) \neq \sigma^2 I_n$$

$$E(x'u) \neq 0$$

u GMM

$$y = X\beta + u \quad u \sim N(0, \sigma^2 I)$$

X is fixed $\text{rank}(X) = k$

$$(x'x) \rightarrow (x'x)^+ \quad b = (x'x)^+ x'y$$

$$b = (x'x)^+ x'y$$

$$= (x'x)^+ x'x\beta + (x'x)^+ x'u$$

$$E(b) = \beta \text{ when } x \text{ is fixed}$$

$$\text{Var}(b) = E[(b - \beta)(b - \beta)']$$

$$= E[(x'x^{-1})' x' u u' x (x'x)^{-1}]$$

$$= \sigma^2 (x'x)^{-1}$$

$$b \sim (\beta, \sigma^2 (x'x)^{-1})$$

BLUE estimator Best Linear Unbiased Estimator
Gauss Markov Theorem.

$$y = x\beta + u$$

$$b_{OLS} = Cy \quad b = \underbrace{(x'x)^{-1} x'}_{k \times n} y$$

$$C = [(x'x)^{-1} x' + A]$$

$$b_{OLS} = Cy$$

$$= [(x'x)^{-1} x' + A] y$$

$$= (x'x)^{-1} x' y + Ay = b + Ay$$

$$b_{\hat{x}} = b + Ay$$

$$= (x'x)^{-1}x'(x\beta + u) + A(x\beta + u) \\ = \beta + (x'x)^{-1}x'u + Ax\beta + Au$$

$$E(b_{\hat{x}}) = E(b) + (x'x)^{-1}x'E(u) + Ax\beta \quad Ax = 0 \\ = \beta$$

$$\text{Var}(b_{\hat{x}}) = E[(b_{\hat{x}} - \beta)(b_{\hat{x}} - \beta)'] \\ = E[(x'x)^{-1}x'u + Au](u'x(x'x)^{-1} + u'A') \\ = E[(x'x)^{-1}x'u u'x(x'x)^{-1} + Au u'x(x'x)^{-1} \\ + (x'x)^{-1}x'u u'A + Au u'A'] \\ = \sigma^2(x'x)^{-1} + \sigma^2 Ax(x'x)^{-1} + \sigma^2 (x'x)^{-1}x'A' + \sigma^2 AA'$$

$$\text{Var}(b_{\hat{x}}) = \sigma^2(x'x)^{-1} + \sigma^2 AA' \quad \text{where } A =$$

$$= \text{Var}(b) + \sigma^2 AA' \\ \hookrightarrow \text{positive definite}$$

$$\text{For any } \text{Var}(b_{\hat{x}}) \succ \text{Var}(b)$$

Geometric Interpretation

$$y = X\beta + u$$

$$y = \mathbb{R}^n$$

$$\dim \ell(x) = k$$

